



Lecture 2 – Static Analysis

30 MCQs (Dr. Ehab Essa Style)

1.

Analyzing code without executing it is called:

- a) Dynamic analysis
- b) Static analysis
- c) Profiling
- d) Debugging

Answer: b) Static analysis

2.

Which technique finds bugs before program execution?

- a) Dynamic checking
- b) Print debugging
- c) Static checking
- d) Profiling

Answer: c) Static checking

3.

Java is classified as a:

- a) Dynamically typed
- b) Weakly typed
- c) Statically typed
- d) Untyped

Answer: c) Statically typed

4.

In static typing, variable types are known at:

- a) Runtime
- b) Execution
- c) Compile time
- d) Deployment

Answer: c) Compile time

5.

Which error is caught before running the program?

- a) Divide by zero
- b) Null dereference
- c) Syntax error
- d) Infinite loop

Answer: c) Syntax error

6.

Divide-by-zero is usually caught by:

- a) Static checking
- b) Dynamic checking
- c) Compilation
- d) Parsing

Answer: b) Dynamic checking

7.

Which language is dynamically typed?

- a) Java
- b) C++
- c) Python
- d) C

Answer: c) Python

8.

Static checking mainly focuses on errors related to:

- a) Values
- b) Types
- c) Performance
- d) Memory

Answer: b) Types

9.

Which error can static checking detect?

- a) Array overflow
- b) Wrong argument type
- c) Divide by zero
- d) Infinite recursion

Answer: b) Wrong argument type

10.

Using code indentation improves:

- a) Compilation
- b) Performance
- c) Readability
- d) Execution

Answer: c) Readability

11.

Which Java type stores actual data directly?

- a) String
- b) Integer
- c) int
- d) List

Answer: c) int

12.

Which Java type stores a reference?

- a) char
- b) boolean
- c) String
- d) double

Answer: c) String

13.

Integer overflow in Java results in:

- a) Compiler error
- b) Runtime exception
- c) Wrapped value
- d) Program stop

Answer: c) Wrapped value

14.

NaN is associated with which type?

- a) int
- b) long
- c) double
- d) char

Answer: c) double

15.

Which tool checks Java bytecode for bug patterns?

- a) JUnit
- b) Checkstyle

- c) FindBugs
- d) Maven

Answer: c) FindBugs

16.

FindBugs classifies bugs by:

- a) Lines
- b) Severity
- c) Speed
- d) Size

Answer: b) Severity

17.

A “Scariest” bug rank belongs to range:

- a) 1–4
- b) 5–9
- c) 10–14
- d) 15–20

Answer: a) 1–4

18.

Which tool focuses on coding style?

- a) JUnit
- b) FindBugs
- c) Checkstyle
- d) Profiler

Answer: c) Checkstyle

19.

Which is NOT a Checkstyle concern?

- a) Indentation
- b) Naming
- c) Javadoc
- d) Algorithm

Answer: d) Algorithm

20.

Arrays in Java are:

- a) Variable length
- b) Immutable length
- c) Dynamic
- d) Resizable

Answer: b) Immutable length

21.

List size is obtained using:

- a) length()
- b) count()
- c) size()
- d) getSize()

Answer: c) size()

22.

Lists store only:

- a) Primitive types
- b) Object types
- c) Functions
- d) Methods

Answer: b) Object types

23.

Which is a wrapper class?

- a) int
- b) double
- c) Integer
- d) char

Answer: c) Integer

24.

A type that cannot change after creation is:

- a) Mutable
- b) Dynamic
- c) Immutable
- d) Flexible

Answer: c) Immutable

25.

Which type is immutable?

- a) ArrayList
- b) StringBuilder
- c) String
- d) List

Answer: c) String

26.

Which keyword makes a reference immutable?

- a) static
- b) const
- c) final
- d) public

Answer: c) final

27.

Aliasing occurs when:

- a) Two methods exist
- b) Two references point same object
- c) Object is immutable
- d) Code is duplicated

Answer: b) Two references point same object

28.

Mutability increases risk mainly because of:

- a) Speed
- b) Aliasing
- c) Compilation
- d) Syntax

Answer: b) Aliasing

29.

Changing the content of a mutable object is called:

- a) Reassignment
- b) Mutation
- c) Compilation
- d) Casting

Answer: b) Mutation

30.

Static analysis improves software mainly by:

- a) Running faster
- b) Early bug detection
- c) Better UI
- d) Shorter code

Answer: b) Early bug detection

Lecture 2 – Static Analysis

Essay Questions with Model Answers

Essay Question 1

Explain the difference between static checking and dynamic checking.

Model Answer:

Static checking analyzes the program **before execution**.

It is mainly used to detect **type errors**, **syntax errors**, and **incompatible operations**.

Static checking is fast and can find many bugs **early**, but it cannot detect errors that depend on **runtime values**.

Dynamic checking analyzes the program **during execution**.

It detects errors such as **division by zero**, **null pointer exceptions**, and **array bounds violations**.

Dynamic checking can find errors missed by static checking but requires the program to run.

Essay Question 2

Discuss the role of mutability and immutability in software correctness.

Model Answer:

Mutability means that an object's state **can change after creation**.

Mutable objects increase the risk of **aliasing**, where multiple references point to the same object, leading to **unexpected side effects**.

Immutability means that an object's state **cannot change after creation**.

Immutable objects are safer, easier to reason about, and help improve **program correctness**.

Examples of immutable types in Java include **String** and wrapper classes.



Essay Question 3

Why is static analysis important in software construction? Explain its benefits and limitations.

Model Answer:

Static analysis is important because it finds **bugs before program execution**. It improves **software quality**, **correctness**, and **maintainability**.

Benefits include:

- Early detection of errors
- No need to run the program
- Improved code readability using tools like **Checkstyle**

Limitations include:

- Cannot detect runtime-dependent errors
- May produce false positives
- Cannot replace dynamic testing